SAT Math Mastery · Session 9

Right Triangles, Trigonometry and Circles

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics right-triangle trigonometry (sin, cos, tan and cofunctions) · special right triangles and the Pythagorean theorem · altitude to the hypotenuse · Laws of Sines and Cosines, area with sin · distance and applied right triangles · circle equations (completing the square) · arcs, chords, tangents and inscribed angles · radians and degrees

Name / Date _____ _____

PART A · IN CLASS — 36 PROBLEMS

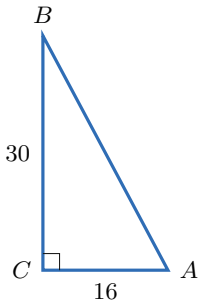
Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). In the Answers PDF, grey boxes show exactly what to type in Desmos; tap a grey dot to read an intersection, intercept or point. Type deg after an angle for degrees (for example sin(35 deg)); a letter such as k or b becomes a slider when Desmos offers one.

1. Medium · Desmos

If $\sin(2x)^\circ = \cos(x+15)^\circ$ and $0 < x < 45$, what is the value of x?

2. Medium · No calculator needed

In right triangle ABC, the right angle is at C. If $AC = 16$ and $BC = 30$, what is $\tan A$?



(A) $\frac{16}{34}$

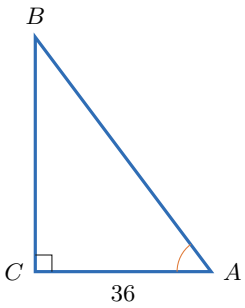
(B) $\frac{34}{16}$

(C) $\frac{8}{15}$

(D) $\frac{15}{8}$

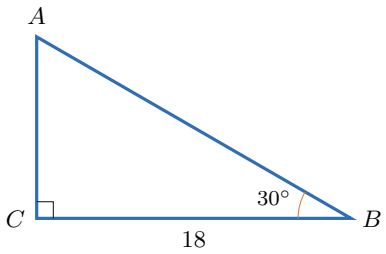
3. HARD · DESMOS

In right triangle ABC, the right angle is at C, and $\cos A = \dfrac{3}{5}$. If $AC = 36$, what is the length of AB?



4. MEDIUM · DESMOS

In triangle ABC, angle $C = 90°$, angle $B = 30°$, and $BC = 18$. What is the length of AC?



(A) 9

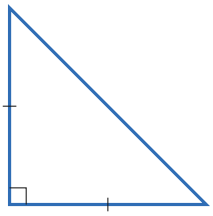
(B) $6\sqrt{3}$

(C) $9\sqrt{3}$

(D) $18\sqrt{3}$

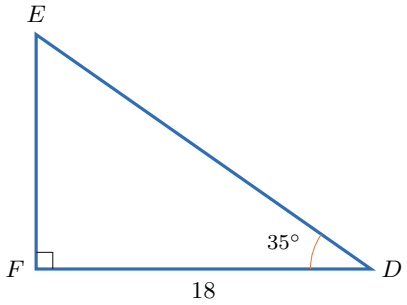
5. HARD · DESMOS

An isosceles right triangle has a perimeter of $40 + 20\sqrt{2}$. What is the length of the hypotenuse of the triangle?



6. HARD · DESMOS

In right triangle DEF, angle F is the right angle, angle D measures 35°, and side DF (adjacent to angle D) measures 18. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 10.3

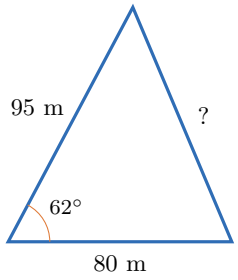
(B) 12.6

(C) 14.9

(D) 17.1

7. Hard · Desmos

A triangular plot of land has two sides measuring 80 meters and 95 meters, with an included angle of 62° between them. Which of the following is closest to the length, in meters, of the third side?



(A) 83.4

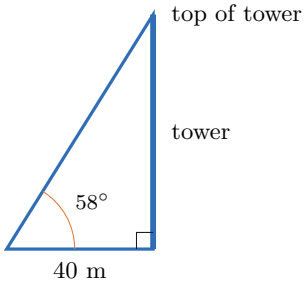
(B) 87.9

(C) 91.0

(D) 95.7

8. Hard · Desmos

From a point on the ground 40 meters from the base of a communications tower, the angle of elevation to the top of the tower is 58°. Which of the following is closest to the height, in meters, of the tower?



(A) 33.9

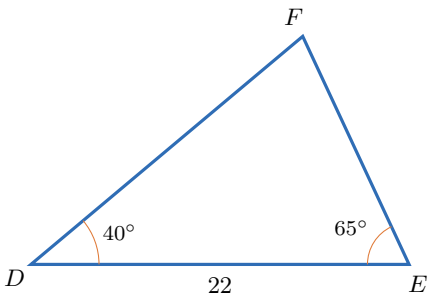
(B) 47.1

(C) 64.0

(D) 75.5

9. HARD · DESMOS

In triangle DEF, angle $D = 40°$, angle $E = 65°$, and side $DE = 22$. Which of the following is closest to the length of side EF (opposite angle D)?



(A) 12.6

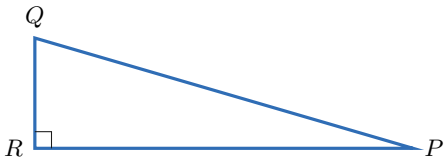
(B) 14.6

(C) 16.9

(D) 19.4

10. HARD · NO CALCULATOR NEEDED

In right triangle PQR, angle R is the right angle. If $\sin(P) = \dfrac{7}{25}$, what is the value of $\cos(Q)$?



(A) $\dfrac{7}{25}$

(B) $\dfrac{24}{25}$

(C) $\dfrac{25}{7}$

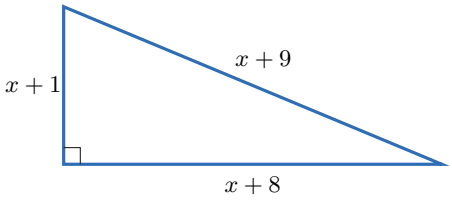
(D) $\dfrac{25}{24}$

11. HARD · NO CALCULATOR NEEDED

Maya walked from Noah's house 4 miles due north, then 3 miles due east, while Noah walked 2 miles due south and 5 miles due west. If Noah were to walk directly to Maya's location, how many hours would it take him, if Noah walks at a rate of 5 miles per hour?

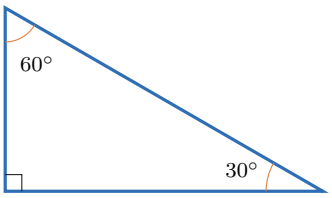
12. HARD · DESMOS

The legs of a right triangle have lengths $x+1$ and $x+8$, and the hypotenuse has length $x+9$. What is the area of the triangle?



13. HARD · DESMOS

A 30°-60°-90° triangle has a perimeter of $18 + 6\sqrt{3}$. What is the length of its hypotenuse?



(A) 6

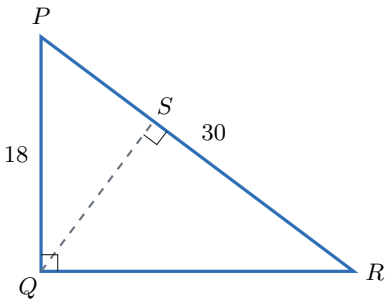
(B) $6\sqrt{3}$

(C) 12

(D) $12\sqrt{3}$

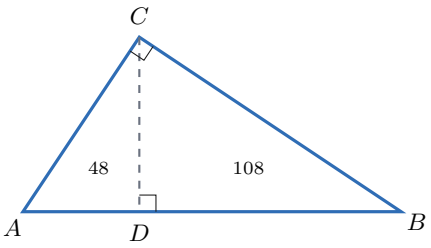
14. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 18$, and $PR = 30$. Point S lies on $\overline{PR}$ such that $\overline{QS} \perp \overline{PR}$. What is the value of $\dfrac{QR}{QS}$?



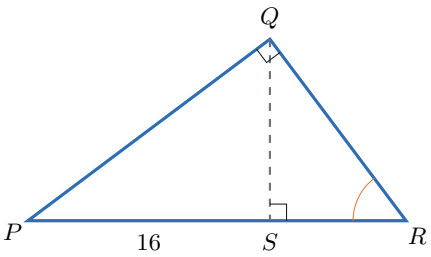
15. HARD · DESMOS

In triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?



16. HARD · DESMOS

In triangle PQR, the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan\angle QRP = \dfrac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?



(A) 12

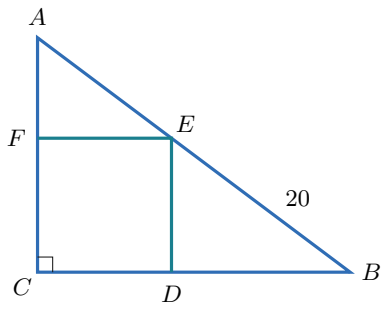
(B) 15

(C) 25

(D) 9

17. HARD · DESMOS

In right triangle ABC, $\angle C = 90°$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



(A) $\frac{48}{7}$

(B) $\frac{24}{7}$

(C) $\frac{48}{5}$

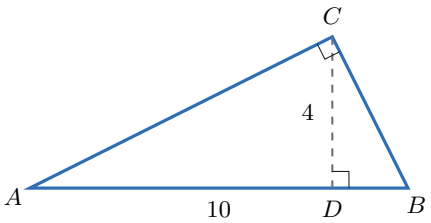
(D) $\frac{96}{7}$

18. HARD · DESMOS

In right triangle PQR, $\angle R = 90°$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

19. HARD · DESMOS

In right triangle ABC, the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?



(A) $\dfrac{\sqrt{5}}{5}$

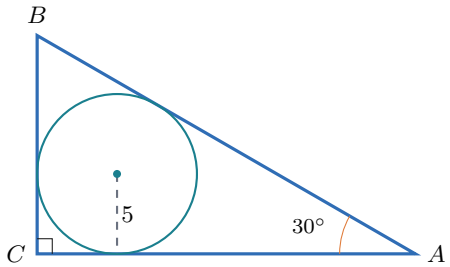
(B) $\dfrac{4}{5}$

(C) $\dfrac{1}{2}$

(D) $\dfrac{2\sqrt{5}}{5}$

20. HARD · DESMOS

In triangle ABC, angle C is a right angle and angle A measures $30°$. The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?



(A) $15 + 5\sqrt{3}$

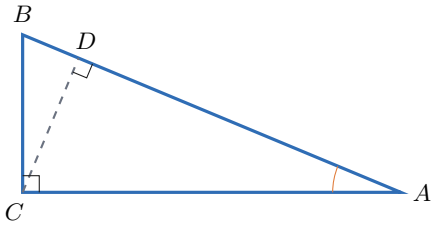
(B) $10 + 10\sqrt{3}$

(C) $10\sqrt{3} - 10$

(D) $5 + 5\sqrt{3}$

21. HARD · DESMOS

In right triangle ABC, the right angle is at C and $\tan A = \dfrac{5}{12}$. The area of the triangle is 120. Point D lies on $\overline{AB}$ such that $\overline{CD} \perp \overline{AB}$. What is the length of $\overline{AD}$?



(A) $\dfrac{120}{13}$

(B) 24

(C) $\dfrac{50}{13}$

(D) $\dfrac{288}{13}$

22. HARD · DESMOS

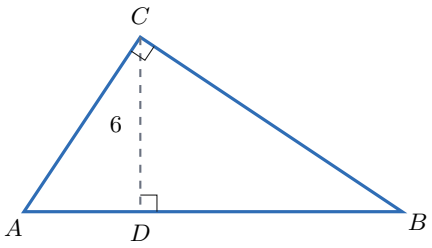
For $0 < x < 45$, the equation

$$\sin\left((5x - 20)^\circ\right) = \cos\left((x + 14)^\circ\right)$$

has more than one solution. What is the sum of all solutions x in this interval?

23. HARD · DESMOS

In right triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD. What is the value of $\sin B$?



(A) $\dfrac{2\sqrt{13}}{13}$

(B) $\dfrac{3\sqrt{13}}{13}$

(C) $\dfrac{4}{13}$

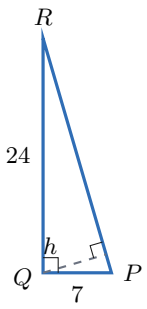
(D) $\dfrac{6}{13}$

24. HARD · DESMOS

In triangle PQR, the measure of angle P is $30°$ and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

25. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 7$ and $QR = 24$. The altitude from Q to hypotenuse $\overline{PR}$ has length h. What is the value of h? (Enter as a decimal.)



26. HARD · DESMOS

In a right triangle, angle θ is acute and $\sin\theta = 0.6$. What is the value of $\cos(90° - \theta) + \tan\theta$?

(A) 1.35

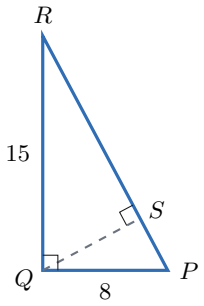
(B) 1.6

(C) 0.75

(D) 1.2

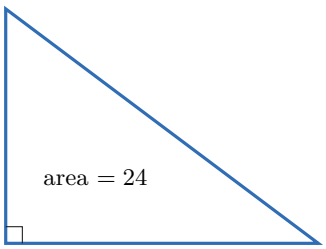
27. HARD · DESMOS

In right triangle PQR, the right angle is at Q, $PQ = 8$ and $QR = 15$. Point S is on $\overline{PR}$ such that $\overline{QS}$ is perpendicular to $\overline{PR}$. What is the length of $\overline{QS}$? (Enter as a fraction.)



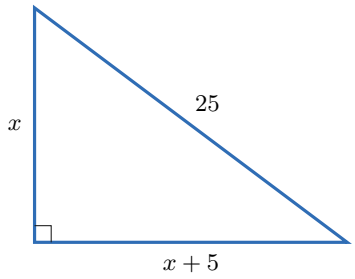
28. HARD · DESMOS

In a right triangle, the sine of one acute angle is $\frac{3}{5}$ and the area of the triangle is 24. What is the length of the hypotenuse?



29. HARD · DESMOS

The legs of a right triangle have lengths x and $x + 5$, and the hypotenuse is 25. What is the area of the triangle?

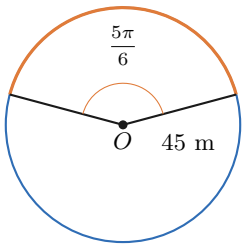


30. HARD · DESMOS

In the xy-plane, the graph of $x^2 + y^2 + 4x - 10y + k = 0$ is a circle with radius 3. What is the value of k?

31. HARD · DESMOS

A circular plaza has a radius of 45 meters. A decorative walking path forms an arc along the plaza's boundary that subtends a central angle of $\frac{5\pi}{6}$ radians. What is the length of the arc, in meters?



(A) $\frac{25\pi}{2}$

(B) $\frac{75\pi}{2}$

(C) 45π

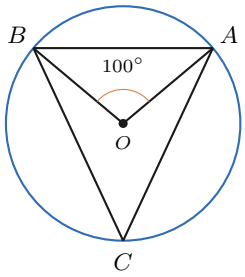
(D) 150π

32. HARD · DESMOS

A circle in the xy-plane has the equation $x^2 + y^2 - 8x + 2y - 19 = 0$. What is the diameter of the circle?

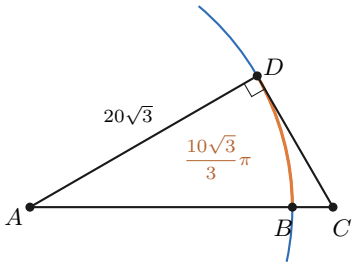
33. HARD · DESMOS

In the circle above, chord AB subtends a central angle of $100°$ at center O. Point C lies on the major arc. What is the measure of angle ACB, in degrees?



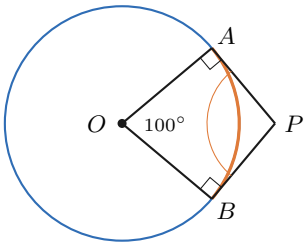
34. HARD · DESMOS

A circle has center A and radius $20\sqrt{3}$. Segment CD is tangent to the circle at point D, so triangle ACD has a right angle at D. Point B lies on the circle along ray AC, and minor arc BD has length $\dfrac{10\sqrt{3}}{3}\pi$. What is the length of CD?



35. HARD · DESMOS

In a figure, PA and PB are tangent to a circle centered at O, at points A and B respectively, so $OA \perp PA$ and $OB \perp PB$. If $\angle APB = 100°$ and the circle has a circumference of 36π, what is the length of minor arc AB?



(A) 6π

(B) 8π

(C) 9π

(D) 10π

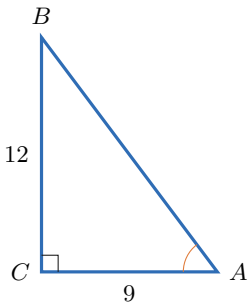
36. HARD · DESMOS

A circle has equation $(x+4)^2 + (y-b)^2 = 100$, where b is a constant. The points $(4, 13)$ and $(-10, -1)$ both lie on the circle. What is the value of b?

PART B · SELF-ASSESSMENT — 21 PROBLEMS

37. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AC = 9$, and $BC = 12$. What is $\sin(A)$?



(A) $\frac{4}{5}$

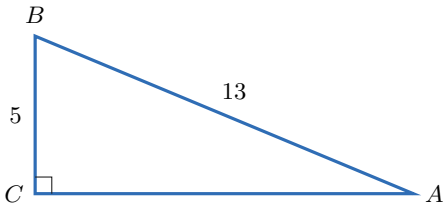
(B) $\frac{3}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{5}$

38. MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC, angle C is the right angle, $AB = 13$ (the hypotenuse), and $BC = 5$. What is $\cos(A)$?



(A) $\frac{5}{12}$

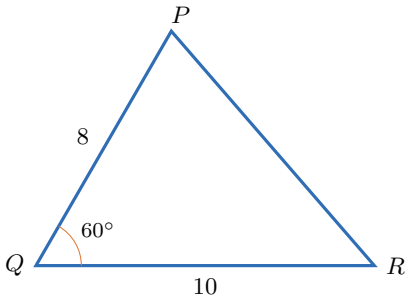
(B) $\frac{12}{5}$

(C) $\frac{5}{13}$

(D) $\frac{12}{13}$

39. Medium · Desmos

In $\triangle PQR$, $PQ = 8$, $QR = 10$, and $m\angle Q = 60°$. What is the area of $\triangle PQR$?



(A) 20

(B) $20\sqrt{2}$

(C) $20\sqrt{3}$

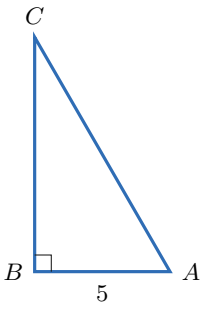
(D) $40\sqrt{3}$

40. Medium · No calculator needed

What is the distance between the points $(-3, -9)$ and $(9, -4)$?

41. Medium · Desmos

In triangle ABC, angle B is a right angle and the measure of angle A is twice the measure of angle C. If $AB = 5$, what is the length of BC?



(A) 5

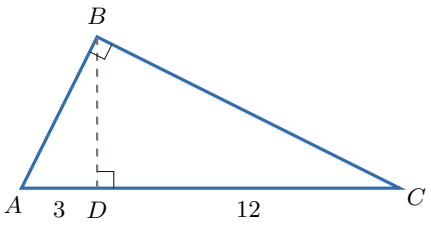
(B) $5\sqrt{2}$

(C) $5\sqrt{3}$

(D) 10

42. Medium · Desmos

In right triangle ABC, the right angle is at B, and BD is the altitude drawn to hypotenuse AC (with D on AC). If $AD = 3$ and $DC = 12$, what is the length of BD?



(A) 4

(B) 6

(C) 7.5

(D) 9

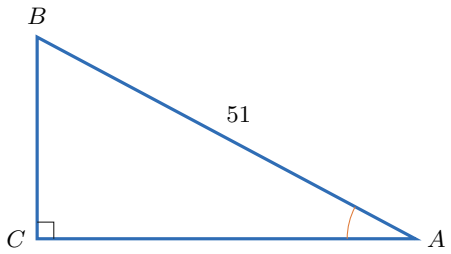
43. Medium · Desmos

$$\sin(x^\circ) = \cos\left((2x+9)^\circ\right)$$

In the equation above, x is between 0 and 30. What is the value of x?

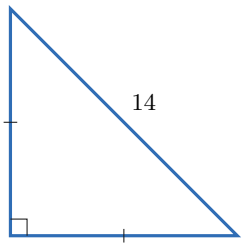
44. Medium · Desmos

In right triangle ABC, the right angle is at C, $\tan A = \dfrac{8}{15}$, and $AB = 51$. What is the perimeter of triangle ABC?



45. Medium · Desmos

An isosceles right triangle has a hypotenuse of length 14. What is the perimeter of the triangle?



(A) $14 + 7\sqrt{3}$

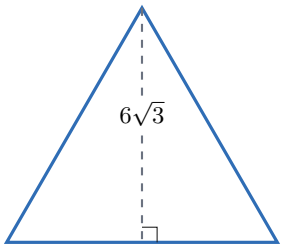
(B) 28

(C) $14 + 7\sqrt{2}$

(D) $14 + 14\sqrt{2}$

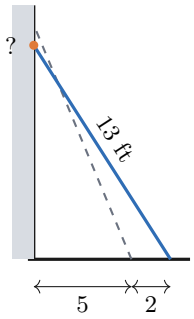
46. Medium · Desmos

The height of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?



47. Medium · Desmos

A 13-foot ladder leans against a vertical wall. Its base is 5 feet from the wall. The base is then pulled 2 feet farther from the wall, so the ladder is 7 feet from the wall. To the nearest tenth of a foot, how high up the wall does the ladder reach now?



48. Medium · No calculator needed

What is the value of $\cos\dfrac{5\pi}{3}$?

(A) $-\dfrac{\sqrt{3}}{2}$

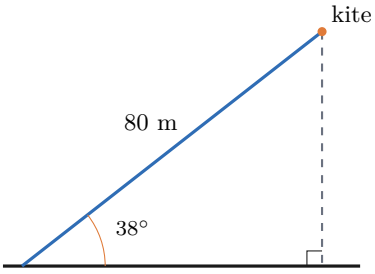
(B) $-\dfrac{1}{2}$

(C) $\dfrac{\sqrt{3}}{2}$

(D) $\dfrac{1}{2}$

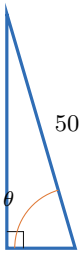
49. Medium · Desmos

A kite is flying at the end of a straight string that is 80 meters long. The string makes an angle of 38° with level ground. To the nearest meter, how high above the ground is the kite?



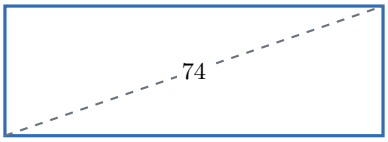
50. Medium · Desmos

In a right triangle, the hypotenuse is 50 and the cosine of an acute angle θ is $\frac{7}{25}$. What is the area of the triangle?



51. Medium · Desmos

The diagonal of a rectangle is 74 and the ratio of its two side lengths is $12 : 35$. What is the area of the rectangle?



52. Medium · Desmos

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of the circle?

(A) 3

(B) 4

(C) 5

(D) 25

53. Medium · Desmos

A circle in the xy-plane is defined by the equation $x^2 + y^2 + 10x - 4y + 13 = 0$. What is the radius of the circle?

(A) 4

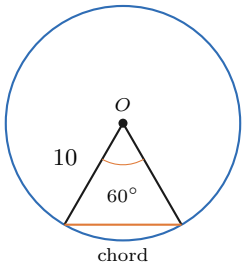
(B) 8

(C) 13

(D) 16

54. Medium · Desmos

In a circle with radius 10, a chord subtends a central angle of 60°. What is the length of the chord?



(A) 5

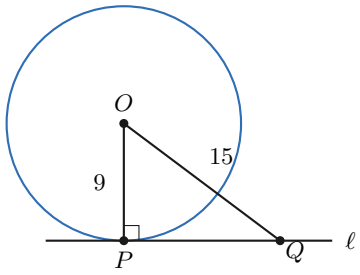
(B) $5\sqrt{3}$

(C) 10

(D) $10\sqrt{3}$

55. Medium · Desmos

In the figure, line ℓ is tangent to the circle with center O at point P. If $OP = 9$ and $OQ = 15$ for a point Q on line ℓ, what is the length of PQ?



(A) 6

(B) 12

(C) $\sqrt{306}$

(D) 18

56. Medium · No calculator needed

An angle has a measure of $\dfrac{7\pi}{12}$ radians. What is the measure of the angle, in degrees?

57. Medium · No calculator needed

The number of radians in a 540-degree angle can be written as $a\pi$, where a is a constant. What is the value of a?